

change and end of lifetime), we find that there is a 24% chance that the risk reductions for a 55-year-old occur between ages 55 and 64, a 37% chance that the case reductions occurs between ages 65 and 74, etc. For distributing avoided cases within an age bin, we assume an equal incidence distribution across years within each bin.

Table H-11. Percent Lung and Bronchus Cancer Incidence by Age and Distribution of Risk Reduction by Age for an Exposure Change at 55

Age Group	Percent New Cases per Year by Age*	Percent of New Cases Occurring at or After Age 55 ¹
0-20	0	NA
20-34	0.2	NA
35-44	0.9	NA
45-54	6.6	NA
55-64	21.8	24
65-74	34.1	37
75-84	26.6	29
85-99	9.7	11
55-99	92.2	100

*May not sum to 100% due to rounding

¹Calculated as the percentage in column 2 divided by 92.2%, where 92.2% is the percentage of lung and bronchus incidence between age 55 and 99.

H.3.2 Out of Hospital Cardiac Arrest

The COI for cardiac arrests occurring outside of the hospital is derived from O'Sullivan et al. (2011), who estimate three-year medical costs associated with cardiovascular disease events among adults ages 35 and older in the U.S. The authors rely on administrative claims data from a large U.S. health plan and develop econometric models to predict medical costs for 15 different cardiovascular events, including cardiac arrest, referred to as resuscitated cardiac arrest. The dataset includes over 20 million commercial and Medical Advantage members between 2002 and 2006. Cardiac arrests are identified using the ICD-9 code 427.5. The authors use propensity score matching to develop a control group with which to compare costs versus individuals that suffered cardiac arrest. Medical costs occurring within the month of the event were excluded to avoid double counting hospitalization costs, which are separately captured by the hospitalization valuation functions. Over three years, the total medical costs, excluding hospitalization, are \$36,142 (undiscounted, inflated to 2015\$), or \$35,880 using a 2% discount rate, \$35,753 using a 3% discount rate and \$35,282 for a 7% discount rate (Table H-12).

Table H-12. Valuation Estimate for Cardiac Arrests (2015\$)

Costs	Cumulative Costs	Annual Costs			
		Undiscounted	2% Discount Rate	3% Discount Rate	7% Discount Rate
<i>Month of Event*</i>	\$43,904	\$43,904	\$43,904	\$43,904	\$43,904
Year 1	\$71,901	\$27,997	\$27,997	\$27,997	\$27,997
Year 2	\$74,701	\$2,800	\$2,745	\$2,718	\$2,617
Year 3	\$80,046	\$5,345	\$5,138	\$5,038	\$4,668
Years 1-3	\$80,046	\$36,142	\$35,880	\$35,753	\$35,282

*Excluded to avoid double-counting with hospitalization costs.

H.3.3 Stroke

Maniloff and Fann (2023) developed COI estimates of direct and delayed health care costs, loss of wage income, and home paid caregiving costs associated with a non-fatal first stroke. The authors conduct a regression analysis based on 2017 – 2019 medical expenditure data from the Agency for Healthcare Research and Quality’s (AHRQ’s) Medical Expenditure Panel Survey (MEPS) to estimate the difference in total medical expenditures between individuals who report having been diagnosed with new onset stroke and otherwise similar individuals who do report a stroke diagnosis. This approach allows for inclusion of direct medical costs due to the stroke itself and medical costs due subsequent health effects that may be related to having experienced a stroke (sequelae). The MEPS data includes actual (not billed) insurer payments, government payments and individual payments, and includes agency and non-agency paid caregiving⁴² in its caregiving cost estimates. The authors report the annual costs and present value of lifetime costs of a first stroke for those ages 18 and older, broken into three cost categories: health care, lost wage income, and home caregiving. The authors estimated lifetime costs using data on expected remaining life expectancy at age of first stroke; the resulting present value lifetime estimates are the values used in the BenMAP valuation function. The study estimated the average present value lifetime COI for a new onset stroke to be \$159,067 (2015\$) using a 2% discount rate, \$150,675 (2015\$) using a 3% discount rate and \$124,845 (2015\$) using a 7% discount rate.

H.4 Acute Symptoms and Illness Not Requiring Hospitalization

Several acute symptoms and illnesses have been associated with air pollution, including acute bronchitis in children, upper and lower respiratory symptoms, and exacerbation of asthma (as indicated by one of several symptoms whose occurrence in an asthmatic generally suggests the onset of an asthma episode). In addition, several more general health endpoints which are associated with one or more of these acute symptoms and illnesses, such as minor restricted activity days, school loss days, and work loss days, have also been associated with air pollution. We briefly discuss the derivation of the

⁴² Agency caregivers are individuals who are employed by an established home care agency, whereas non-agency workers include private and independent at-home caregivers.

unit values for acute respiratory symptoms (minor restricted activity days), asthma exacerbation, and school loss days. Tables H-13 and H-14 summarize the values used by EPA in their regulatory impact analyses.

Table H-13. Additional Unit Values Available for Myocardial Infarction

Basis of Estimate	Age Range		Medical Cost (2015\$)	Opportunity Cost (2015\$)	Total Cost (2015\$)
	Min	Max			
COI: present value of medical and opportunity wage costs over estimated years of life at the time of diagnosis, 2% DR, Maniloff & Fann (2023)	18	44	\$167,000	\$368,000 (\$139,000 - \$1,100,000)	\$539,000 (\$306,000 - \$1,300,000)
	45	54	\$83,000	\$218,000 (\$82,000 - \$674,000)	\$301,000 (\$166,000 - \$757,000)
	55	64	\$57,000	\$148,000 (\$56,000 - \$456,000)	\$204,000 (\$112,000 - \$512,000)
	65	99	\$45,000	\$86,000 (\$32,000 - \$265,000)	\$131,000 (\$78,000 - \$311,000)
COI: present value of medical and opportunity wage costs over estimated years of life at the time of diagnosis, 3% DR, Maniloff & Fann (2023)	18	44	\$151,000	\$334,000 (\$126,000 - \$1,033,000)	\$485,000 (\$277,000 - \$1,184,000)
	45	54	\$78,000	\$203,000 (\$77,000 - \$628,000)	\$281,000 (\$155,000 - \$706,000)
	55	64	\$53,000	\$140,000 (\$53,000 - \$432,000)	\$193,000 (\$106,000 - \$485,000)
	65	99	\$44,000	\$83,000 (\$31,000 - \$255,000)	\$127,000 (\$75,000 - \$299,000)
COI: present value of medical and opportunity wage costs over estimated years of life at the time of diagnosis, 7% DR, Maniloff & Fann (2023)	18	44	\$108,000	\$238,000 (\$90,000 - \$734,000)	\$346,000 (\$198,000 - \$842,000)
	45	54	\$61,000	\$160,000 (\$60,000 - \$495,000)	\$222,000 (\$122,000 - \$556,000)
	55	64	\$44,000	\$114,000 (\$43,000 - \$353,000)	\$158,000 (\$87,000 - \$397,000)
	65	99	\$38,000	\$72,000 (\$27,000 - \$224,000)	\$111,000 (\$66,000 - \$262,000)

* Costs are rounded to the nearest thousand dollars. Because we scale the wage loss estimates from Maniloff & Fann (2023) using county-level workers' net costs and labor force participation rates, the values for the opportunity wage costs are not constant and vary by county. Consequently, the total costs also vary by county. For the opportunity costs and total

costs, we present the average costs across all counties and a range with the minimum and maximum values (in parentheses).

Table H-14. Core Unit Values Available for Acute Symptoms and Illnesses

Health Endpoint	Basis for Estimate	Age Range		Unit Value (2015\$)	Distribution of Unit Value	Parameters of Distribution	
		Min	Max			P1	P2
Minor Restricted Activity Days	WTP: 1 day, CV studies	18	99	\$70	Triangular	28.51	110.62
New Onset Asthma	COI: lifetime med, lifetime productivity, 2% DR	18	99	\$182,681	None	0	0
	COI: lifetime med, lifetime productivity, 3% DR	18	99	\$146,370	None	0	0
	COI: lifetime med, lifetime productivity, 7% DR	18	99	\$77,490	None	0	0
Cough; Chest Tightness; Shortness of Breath; Wheeze	WTP: 1 symptom-day, Dickie and Mesmen (2004)	0	17	\$219	LogNormal	5.390	0.078
		18	99	\$115	LogNormal	5.390	0.078
Asthma Symptoms (Albuterol Use), Albuterol Use	COI: use of inhaler	0	99	\$0.35	None	0	0
Allergic Rhinitis	COI: 1 yr med costs	0	17	\$600	None	0	0
Work Loss Days *	County average of mean daily wage	18	65	\$298*	None	N/A	N/A
School Loss Days	County average of caregiver costs and loss of learning, 2% DR	5	6	\$4,056*	None	N/A	N/A
		7	12	\$1,235*	None	N/A	N/A
		13	18	\$903*	None	N/A	N/A
	County average of caregiver costs and loss of learning, 3% DR	5	6	\$3,070*	None	N/A	N/A
		7	12	\$977*	None	N/A	N/A
		13	18	\$731*	None	N/A	N/A
	County average of caregiver costs and loss of learning, 7% DR	5	6	\$1,265*	None	N/A	N/A
		7	12	\$506*	None	N/A	N/A
		13	18	\$417*	None	N/A	N/A